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A deep learning based architecture for multi-class skin cancer classification

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Abstract
One of the deadliest forms of skin cancer is malignant melanoma, developed by aberrant melanocyte cell development. Efficient diagnostic procedures are essential due to the rising prevalence of skin illnesses resulting from several variables such as infections, allergies, dietary habits, chemical exposure, genetic sensitivities, and lifestyle changes. An increasing number of Deep Convolutional Neural Network (DCNN) prototypes are used for the computerized identification of skin cancer using dermoscopic images. The research aims to develop a detailed classification framework for the multi-class categorization of skin cancer carcinoma at an earlier phase using Hair removal image processing Technique and Ensemble Visual Geometry Group-16 (EVGG-16) based on a Convolution Neural Network. The main idea is to set different initializations to the same VGG-16 model three times and then to Ensemble the results. The Experimental results show that the VGG-16 model receives an average accuracy of 87% and an F1 score of 86%. The proposed Ensemble Visual Geometry Group-16 (EVGG-16) model receives an average accuracy of 88% and an F1-score of 88% without the hair removal image processing technique; the same model receives an average accuracy of 89% and an F1-score of 88% after applying the hair removal image processing technique. The publicly accessible HAM1000 dataset was employed to experiment; it includes processed images belonging to seven classes to evaluate the model's effectiveness. In conclusion, our proposed Ensemble model with Hair removal technique performs better than other cutting-edge deep-learning techniques and skilled dermatologists. Furthermore, this might be a trustworthy real-time teaching resource for medical students in a dermatology program.

Keywords VGG16 · Skin cancer · Ensemble Learning · Hair Removal Technique

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1 Introduction

Artificial Intelligence is the field of modern computer science that simulates human brain functions, generally problem-solving and learning. Machine learning (ML) is a subfield of Artificial Intelligence (AI) or an application that makes a machine intelligent without being directly programmed and earns experience or performance over time. Deep learning (DL) is a subfield of ML, also known as deep neural networks or deep neural learning. DL uses the underlying neural network architecture known as deep neural networks. Biological neurons inspire neural networks, the basic structure in human and animal brains. Neural dendrites are essential for several cognitive processes, such as perception, sensory processing, spatial navigation, and adaptive learning [1]. The use of brain-inspired computer techniques offers a viable means of accelerating the advancement of artificial general intelligence (AGI) [2] and spiking neural networks (SNNs) [3, 4]. DL applications include object detection [5], healthcare [6–8], chatbots, entertainment, image captioning, etc.

DL has gained immense popularity in healthcare for several reasons, including improved diagnostics, cancer detection, virtual health assistance, etc. Melanoma, or malignant melanoma, is among the most prominent lethal types of skin malignancy. It is typically brought on by extended exposure to ultraviolet (UV) light, resulting in mutations in the melanocyte cells containing pigment called melanin [9]. Skin cancer develops from the DNA changes that cells undergo, resulting from prolonged exposure to UV light [10]. When UV rays reach melanocytes, they induce an overabundance of melanin to be secreted, which results in dark-colored moles. These skin lesions or moles exposed may eventually grow out of control, leading to malignant tumors [11], and rapidly propagate to other body parts.

Health is a major concern as it directly affects quality of life and overall well-being. Addressing health issues is crucial for preventing disease and ensuring long-term vitality [12, 13]. According to statements by the World Health Organisation, one-third of all cancer patients analyzed worldwide are skin cancer cases [14]. An estimated 5.4 million new cases of skin cancer are reported in the US each year, making skin cancer a worldwide public health concern [15]. However, melanomas account for around 74% of all skin cancer-related mortality, with more than 10,000 deaths happening in the US alone each year. Every year, more than a million newly diagnosed instances of melanoma are reported in Europe [16]. Additionally, previous decades have seen an increase in the incidence rates of skin cancer. Melanoma incidence has increased by 119% in the UK since the 1990s or 250% in the US [17]. The depletion of the ozone layer and using tanning beds and solariums are responsible for this trend.

In the past, skin cancer was detected through physical examination and biopsy. While biopsies are among the most convenient methods to identify skin cancer, the procedure is laborious and unpredictable. Macroscopic and dermoscopic skin images have become the most widely used non-invasive tools to help dermatologists diagnose skin cancer [18]. Visible images captured with cameras or mobile phones result in inferior quality and resolution [19]. High-resolution skin images called dermoscopy images for more accurate skin cancer diagnosis [20].

Dermatologists find it difficult to diagnose skin cancer accurately, even when using dermoscopy images because different forms of skin cancer might have similar initial impressions. Furthermore, because experienced dermatologists are exposed to a small portion of the potential cases of skin cancer in their lifetime, their knowledge is restricted to their studies and experiences. Dermatologists typically diagnose skin cancer with an accuracy rate of 62% to 80% [21]. Clinical dermatologists with three to five years of experience have

a 62% diagnosis accuracy rate in reports on their accuracy. However, dermatologists with over ten years of expertise can attain 80% accuracy. For dermatologists with less experience, the performance declined substantially.

The primary disadvantage of dermoscopy is the need for substantial training. This field of research has made great strides toward creating computer-aided diagnostic systems to address the problems dermatologists encounter [22]. A growing amount of data surfaces improves computer-aided diagnostics. It is easy to retrain the system using new data, and the underlying prototype has been expanded to incorporate a wealth of additional medical data into its channel for prediction. The exceptional ability of deep convolutional neural networks to categorize images has led to their application in the medical domain, including in categorizing skin cancer images [23].

The article aims to introduce the state-of-the-art method for accurately multi-classifying skin cancer from dermoscopic images (HAM1000 dataset) using ensemble and image processing techniques. The model's usefulness is to use a dermoscopic image of the skin's afflicted area to identify the class type of skin cancer. The focus of the study is to improve the accuracy of classification. Hair removal is essential for accurate classification results, especially in the early stages of the disease. Convolution Neural Network can more easily assess the skin's surface and spot skin cancer anomalies, including changes in skin tone when hair is removed. Thus, it is crucial since the likelihood of successful treatment of skin cancer therapy is significantly increased with early identification.

In this research, we introduced a novel multi-class skin cancer classification method. The following are the main contributions:

- The paper introduces a novel Ensemble VGG16 based architecture for efficient multi-class classification.
- We compare the VGG16 and Ensemble VGG16 Architecture.
- We further processed images based on hair removal techniques.
- We checked the efficiency of the Ensemble VGG16 model using the processed images.
- The method is successfully implemented and tested using a HAM10000 dataset accessible to the general audience. The suggested model performs more accurately for images after the hair removal technique.

The rest of the article is organized as follows. Section 2 describes the related work on recent DL technologies for binary and multi-class skin cancer classification and their accuracies. Section 3 discusses the proposed methodology, including dataset structure, techniques, and model framework. Section 4 describes evaluating metrics, results, and discussion, followed by a conclusion and future work in Section 5.

2 Related works

The classification of melanoma using a convolutional neural network (CNN) was done by Nasr-Esfahani et al. [24]. With 170 training images, the convolutional neural network with two convolution layers attained an average accuracy of 81%. Ali et al. [25] suggested the CNN-based LightNet melanoma classification network. With fewer parameters used for training, the proposed architecture delivered an accuracy of 81.6% on the ISIC archive dataset. The Deep CNN was proposed by Esteva et al. [26] for classifying skin malignancies. The suggested model attained a 72% accuracy after being trained on 129,450

clinical images. Srinivasu et al. [27] proposed MobileNet V2 and Long Short-Term Memory (LSTM), a deep learning-based computerized method, to categorize skin diseases. The proposed model is effective at preserving stateful data for accurate forecasts. Yields a promising result of 85.34% accuracy.

Hosny et al. [28] employed a deep convolution neural network (DCNN) to identify common nevus, atypical nevus, and melanoma from the PH2 skin cancer dataset. Inspired by the many uses of DCNN architecture, the scientists classified different skin cancer classes from the PH2 dataset utilizing AlexNet. Initially, Imagenet's visual recognition was driven by AlexNet's architecture. The architecture of AlexNet was used for Imagenet's visual recognition. Imagenet comprises three fully connected layers, one max-pooling layer, and five convolution layers. There is no pooling layer after the third and fourth convolution layers. The softmax layer replaced the last layer of the AlexNet for better classification. Inthiyaz et al. [29] proposed an automated image-based ML classification technique to identify and classify skin cancer. Image data are analyzed, processed, and relegated using computational methods to consider the attributes of the processed images. Advanced techniques like Convolutional Neural Networks (CNNs) obtain characteristics from an image, the algorithm for the softmax classifier to classify the image, and the result is a diagnostic report. This program performs a better and more trustworthy approach for diagnosing dermatological illnesses than the traditional method because it will be more accurate and produce results quickly.

Ismail et al. [30] intend to examine the effectiveness of pre-trained convolutional neural networks and transfer learning for classifying skin lesion images. It has demonstrated that neural network algorithms have proven helpful for extracting visual features. The framework proposed combining the most effective three pre-trained networks to improve classification accuracy. The trainable convolution layers to this model improved the classification accuracy. 84.01% overall accuracy was attained with the suggested framework. Farooq et al. [31] Proposed Cutting Edge Deep Learning Techniques for Preliminary Stage Skin Cancer Diagnosis. The study was divided into two main parts. In the first phase, images were processed using different filters and hair removal techniques. Deep learning algorithms, including Inception v3 and Mobile Net, were used for training, testing, and validation in the second phase. The final accuracy of 86% was achieved using the methodology.

Ismail et al. [32] used the HAM10000 dataset, which consists of seven types of epidermal lesions, and several pre-trained neural network architectures were trained using the data set. After resizing the images to 224 X 224, data augmentation was performed. Researchers tested several pre-trained deep learning models and found the ResNet50 model to be the most accurate. In the paper, an ensemble approach is proposed to maximize the accuracy. A combined accuracy of 84% was achieved using ResNet50, VGG16, and DenseNet. Lequan et al. [33] proposed an approach that ensures adequate training and learning with limited training data. Using a fully convolutional residual network (FCRN) improves skin lesion segmentation accuracy by integrating contextual information at multiple scales. A two-stage framework is then formed by combining the proposed FCRN for segmentation and other deep residual networks for classification. By segmenting the entire dermoscopy image, the classification network can acquire more meaningful and relevant features, minimizing the scarcity of training data.

Zhang et al. [34] proposed multiple attention residual learning blocks, an average pooling layer, and a classification layer for skin lesion classification in dermoscopy images using attention residual learning convolutional neural networks (ARL-CNNs). It improves its discriminative representation by converging residual learning with attention learning. Instead of using additional layers to learn attention, the proposed method uses the inherent

capability of DCNNs to understand attention. As a result, the ARL-CNN model demonstrated state-of-the-art performance in classifying skin lesions by emphasizing the discriminative parts of lesions adaptively. Chaturvedi et al. [35] proposed an MCS skin cancer classification system with seven categories. Through transfer learning, seven classes of the HAM10000 dataset are trained using a pre-trained MobileNet model. According to the results, classification accuracy was 83.1%, and precision, recall, and F1-score were 89%, 83%, and 83%, respectively.

Yu et al. [36] proposed a CNN-based features extraction method in which 900 images were used to train a pre-trained neural network model, which probably needs to be improved to train a DL algorithm effectively. The accuracy of DRN-50, VGG-16, and GoogleNet was 85.5%, 82.6%, and 84.7%, respectively. Rahman et al. [37] proposed a model based on ensemble learning to categorize seven distinct types of skin lesions per a weighted average. As a base of the ensemble, they used ResNeXt, SeResNeXt, ResNet, Xception, and DenseNet deep neural networks. Average ensembles can significantly improve results. The proposed system works more efficiently than existing systems and can aid dermatologists in their diagnosis procedures.

Perez et al. [38] proposed CNN architectures that perform well on ImageNet when applied to transfer learning tasks. In the ISIC Challenge 2017 dataset, they implemented five splits computed on the 9 CNN architectures, yielding 135 models for melanoma classification. They concluded ensemble learning was imperative after different comparison evaluations for obtaining optimal classification models. Lopez et al. [39] proposed a DL approach to classify a dermoscopic image of a skin lesion as malignant or benign, which addresses the problem of skin lesion classification, particularly early melanoma detection. VGGNet convolutional neural networks are used in the proposed framework to facilitate transfer learning. According to the experimental results, the proposed method achieves a sensitivity value of 78.66% on the ISIC Archive dataset, which is considerably improved than the previous study.

Gouda et al. [40] studied based on the ISIC 2018 dataset, a convolution neural network (CNN) was analyzed to classify malignant and benign tumors. Several transfer learning models, such as Resnet50, InceptionV3, and Inception Resnet, were considered for fine-tuning. ESRGAN is used as a preprocessing step in this study, which is an innovative and unique aspect. Xinyuan et al. [41] proposed DL algorithms to help diagnose four common cutaneous diseases. Using the results of this study, we reviewed the diagnosis/classification scenarios, underlining that both human expertise and computer algorithms contribute to dermatologic diagnosis.

Jayapriya et al. [42] proposed FCNs that can process inputs of almost any size and generate outputs of the corresponding size with efficient inference and learning. A two-stage framework is used to detect melanoma, segment skin lesions, and identify melanoma lesions. Two FCNs are incorporated based on VGG-16 and GoogleNet to improve segmentation accuracy. The framework's performance analysis shows promising accuracy. Alfi et al. [43] present a technique for non-invasively diagnosing melanoma skin cancer that is interpretable and uses DL and ensemble stacking of machine learning models. The findings demonstrate that weighed images of benign and malignant skin moles are in the collection used to train the classifier methods. The findings demonstrate that balanced images of benign and malignant skin moles were used for Transfer learning using pre-trained deep learning models (MobileNet, Xception, ResNet50, ResNet50V2, and DenseNet121) on ImageNet data.

Maniraj et al. [44] proposed a hybrid deep learning (HDL) method that utilizes the sub-band fusion of 3D wavelets. It is a non-intrusive and impartial technique for examining

skin images. The HDL approach's outcomes on the PH2 database images demonstrate that it can efficiently distinguish between normal, benign, and malignant skin images with a mean accuracy of 99.33% and more than 90% sensitivity and specificity. The research presented here supports the finding that HDL can produce better categorization outcomes. Chaturvedi et al. (https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000/data.) used a MobileNet model that was previously trained on about 12,80,000 images from the 2014 ImageNet Challenge. The framework was refined using transfer learning on 10015 dermoscopy images from the HAM10000 dataset. For each dataset's seven categories, the study's simulation yielded a total precision of 83.1%; the top two and top three classes had respective accuracy of 91.36% and 95.34%. Dermatology specialists may find this approach helpful in making decisions at crucial points. Table 1 summarizes related work on skin cancer classification using different DL approaches and datasets.

3 Methodology

In this section, we will discuss datasets used in our study. The hair removal technique for image enhancement. Lastly, we will present the architecture used in our research.

3.1 Dataset

Addressing the fundamental experiments, we employed the Skin Cancer MNIST: HAM10000 dataset [45] comprising processed 10,015 skin cancer images belonging to 7 classes (akiec, bcc, bkl, df, mel, nv, vasc) [35], each of these having a resolution of 450×600. During 20 years, images were acquired from the dermatological department of the "Medical University of Vienna in Austria" and the skin cancer clinic run by "Cliff Rosendahl in Queensland," Australia. Reflectance Confocal Microscopy was used to capture images, and Histopathology, subsequent follow-up, or expert consensus was used to confirm the image labeling. The dataset's metadata reveals only 7,470 dermoscopic skin lesions despite its 10,015 skin lesion images. The remaining 2545 images are duplicates with varying magnifications or viewpoints. Furthermore, as Fig. 1 illustrates, the dataset must be more balanced. For example, there are 6,705 images in the largest class (nv) and only 115 in the smallest class (pdf). Table 2 gives us the distribution of images among seven classes.

3.2 Data preparation

Images underwent a duplicate elimination process to train and test our DL models. The size of our input image is 650×400. The images are resized (224×224) to fit our training model. After that, the test set, which consisted of 828 images, or 15% of the non-duplicate dataset, was prepared for testing. The remaining non-duplicate images of the dataset were split in an 80:20 ratio between the training and validation sets. Figure 2 shows the train, test, and validation data distribution for the ensemble model without hair removal, and Fig. 3 shows the same split, including the hair removal technique.

Table 1 Summary of related work of skin cancer classification using Deep Learning Approaches

S.NO	Work	No. of Classes	Dataset	Method	Results
1	Srinivasu et al. [27]	7	HAM10000	MobileNet V2 + Long Short Term Memory (LSTM)	Accuracy = 85.34%
2	Ismail et al. [32]	7	HAM10000	ResNet50 + VGG16 + DenseNet	Accuracy = 0.84
3	Lequan et al.[34]	2	ISBI 2016	DCNN	AUC = 0.805
4	Zhang et al. [32]	2	ISIC 2017 archive	Residual CNN	Accuracy = 0.85
5	Chaturvedi et al. [35]	7	HAM10000	MobileNet + Transfer Learning	Accuracy = 0.83
6	Yu et al. [36]	2	ISIC archive	VGG-16, GoogleNet	VGG-16 accuracy = 82.6% GoogleNet Accuracy = 84.7%
7	Rahman et al. [37]		HAM10000, ISIC 2019	ResNeXt,DenseNet, seResNeXt,ResNet and Xception	Accuracy = 0.88
8	Perez et al. [38]	2	ISIC 2017	Densenet, InceptionNet, ResNet, VGG19	Accuracy = 0.827
9	Lopez et al.[39]	2	ISIC 2016	VGG16	Accuracy = 0.8133
10	Gouda et al. [40]	2	ISIC 2018	Inception, Resnet50, and InceptionV3	Accuracy Resnet50 (83.7%), InceptionV3 (85.8%), and Inception Resnet (84%)
11	Xinyuan et al.[41]	4	Self dataset	GoogLeNet, Inception V3	Accuracy = 0.8725 for dataset1 Accuracy = 0.8663 for dataset2
12	Jayapriya et al. [42]	2	ISBI 2016	VGG16 and GoogleNet	Accuracy = 88.9
13	Alfi et al. [43]	2	ISIC Archive (2018)	MobileNet, Xception, ResNet50, ResNet50V2, and DenseNet121	Accuracy = 88%
14	Agrahari et al. [44]	7	HAM1000	MobileNet	Top-level categorical accuracy = 80.81%,
15	Chaturvedi et al. (https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000/data)	7	HAM10000	MobileNet	Accuracy = 83.1%

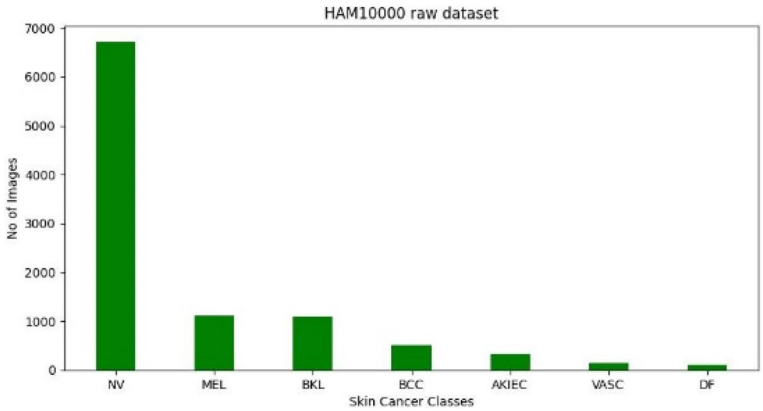


Fig. 1 HAM1000 dataset imbalances

Table 2 HAM1000 dataset class distribution

Class	Df	Vasc	Akiec	Bcc	Bkl	Mel	Nv
No Of Images	115	142	327	514	1099	1113	6705

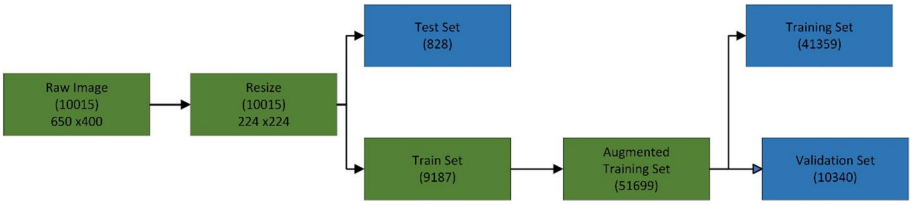


Fig. 2 Train, Validation Test data preparation without hair removal

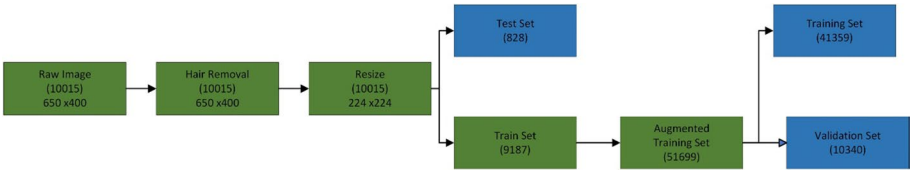


Fig. 3 Train, Validation and Test data preparation with hair removal

3.3 Data Augmentation

Model overfitting is a common challenge when dealing with smaller datasets with DL models [45, 46]. When a model exhibits excellent accuracy on a specific train dataset but produces subpar results on slightly different data called a test dataset, it is said to be overfitting. Therefore, various data augmentation [35, 47] approaches are necessary. Argumentation approaches include flipping horizontally and vertically, rotation, modifying brightness, and adjusting the contrast to expand the training dataset for training

the model. The Keras library provides an ImageDataGenerator function to carry out image augmentation.

The overall quantity of images in the training set is 9078 images, belonging to 7 classes known as vascular lesions(vasc), basal cell carcinoma (bcc), benign keratosis-like lesions (bkl), Actinic keratoses (akiec), melanoma (mel), dermatofibroma (df) and melanocytic nevi (nv) containing 304, 488, 1033, 109,1079, 6042,132 images respectively. Image augmentation is employed to provide the model with unique variants in each training epoch to overcome the constraints of a small dataset. For every epoch, the model Keras' ImageDataGenerator provides augmented training images.

3.4 Hair Removal Technique

Figure 4 shows the hair removal process. The processes that make up each of these are listed below in order of occurrence.

3.4.1 RGB to GRAY transformation process

RGB 650×400×3 images constitute the actual image. Red, green, and blue are the colors present in the image. Transforming this three-dimensional array to a single-dimensional vector is required in many applications, such as fine-line detection. This procedure yields an image that shrinks three channels to one channel. This technique is employed to find tiny hairs on the lesion. The following basic equation is used to apply this strategy.

$$Gray - scale = {}^{(R+G+B)}/_{3}$$

3.4.2 Black-hat transformation process

The Black-hat transformation filter separates delicate hair in the grayscale image. Using this method, regions of intensity of an image in grayscale are located. The grayscale images used in this approach are subjected to morphological processing. Following this, the discrepancies between the initial image and the image produced following the process are analyzed. Binary thresholding excludes pixels of negligible value in the image once calculations to obtain the difference image with hairlines displayed as bright pixels.

3.4.3 The otsu thresholding transformation process

The threshold approach changes images into binary (black and white). It is applied to identify objects and remove image noise [48]. Global thresholding uses the value we choose to do thresholding. By reducing the weight of variation inside the class, the Otsu method attempts to determine the threshold value.

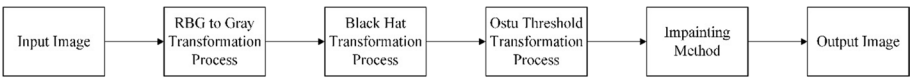


Fig. 4 Hair Removal Process

3.4.4 Inpainting method

Image noise reduction or damage repair is accomplished using in-image painting. These damages are digitally equalized. The defective pixels are swapped out for pixels that resemble nearby pixels. Thus making damaged pixels undetectable.

3.5 Model Architectures

3.5.1 The Architecture of CNN

CNN is a fundamental deep learning model that takes an image as input, allocates weights and trainable biases to various aspects of the image, and differentiates one from the others. During the training phase of the core CNN model, the filters are built manually to acquire knowledge of various training patterns. The CNN model was influenced by the organization of the human brain's visual cortex and shares similarities with the connectivity pattern of architecture neurons. Individual neurons only respond to stimuli within the sensory area, a small portion of the visual field. These fields should overlap to encompass the entire viewing area [49].

VGG-16 Model In an article titled "Very deep convolutional networks for large-scale image recognition," Oxford University researchers Simonyan and Zisserman introduced the CNN-based VGG-16 model [29]. VGG-16, also known as OxfordNet, got its name from the Visual Geometry Group at Oxford University, which enhanced AlexNet by successively substituting big kernel-sized filters 5×5, 11×11 with numerous small 3×3 kernel-sized ones. Based on the VGG-16 network as the feature extraction model, our suggested approach operates due to its outstanding classification performance. Because VGG-16 contains a significantly higher number of nonlinear layers, the network depth is increased, enabling the algorithm to acquire more complex patterns. As a summary, VGG-16 is made up of two convolutional layers: one for feature extraction using a 3×3 convolution kernel size and another for maintaining the key features using a 2×2 max-pooling layer [50]. Every layer in the subsequent convolutional layer contains a convolution kernel size that fits the max-pooling layer's core size. However, there are 64 convolution kernels per convolutional layer. After each max-pooling layer, the number of convolution kernels doubles until it reaches its maximum of 512. To avoid overfitting, two dropout layers, and two 4096-dimensional fully-connected layers are added after the final max-pooling layer. Dropout, which temporarily removes network units based on a specified probability during training, solves these problems. The random gradient descent algorithm eliminates the network components, and every small batch trains a new network. Consequently, this method dramatically enhances the network's universality. The VGG-16 architecture is shown in Fig. 5.

3.5.2 Ensemble Learning

Ensemble learning is a learning process that builds a larger aggregate model by combining multiple basic models that are less capable than the aggregate model [51]. Because there are many different basic models in ensemble learning, the probability of overfitting is decreased. Ensemble learning performs better than individual models and has been

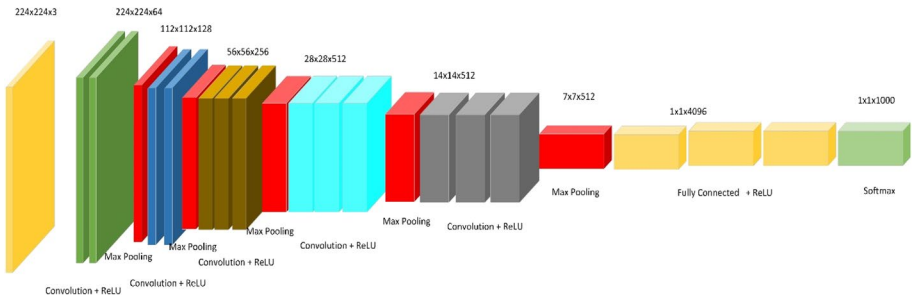


Fig. 5 VGG Architecture

effectively implemented in various fields and instances [52, 53]. Various ensemble strategies differ in how various baseline models are merged and trained. Stacking, random forest, averaging, bagging, and boosting are some of the most popular ensemble approaches [54]. In recent times, the classification of skin lesions has relied on the ensemble technique to attain exact results [55]. Contemporary researchers who use ensemble approaches typically adhere to a similar procedure. An aggregation technique combines the outcomes of many multiclass CNNs trained for a particular job.

The ML community has focused close attention on ensemble learning, a machine learning technique that uses multiple base learners to form ensemble learners for better learning system generalization. Ensemble learning has revealed great success in a variety of artificial intelligence applications. However, as the number of base learners rises, so do the expenses associated with learning multiple base learners and conducting tests using the ensemble learner. Thus, discouraging ensemble learning from being widely used in many artificial intelligence applications. In our experiment, we developed two ensemble models. One ensemble model has Fig. 2 as a train test validation dataset, and another has Fig. 3 as a train test validation dataset. We used the average ensemble learning method in the current study, giving all models identical weights. After that, the average of the Logit outputs from all three models was considered. Logit1, Logit2 and Logit3 is from model1, model 2 and model3 respectively.

$$Prediction = ArgMax\left\{\frac{Logit1 + Logit2 + Logit3}{3}\right\}$$

DL models don't deliver the same outcomes because of differences in their design and complexities; some models perform more successfully than others. Here, we used the same architecture for three models but with varying initialization; due to the randomness of the model, they chose different paths but merged to the same minima. Individual models disagree; that is why the ensemble model performs better.

3.5.3 Setting the Random Seed

Randomness is utilized by machine learning models in both apparent and unexpected ways. Conceptually, this non-determinism could affect the pace of converging in the model, the consistency of the findings, and the network's ultimate quality. It indicates that most likely struggle to replicate the model's outputs across runs, even using the same script and training set. Usually, this seed is selected based on the present time. Nonetheless, we can assign

a value to this random seed to manage the process. In our experiment, we have set three different seed values, 42,52, and 62, for training six models. Two models with similar seed values, one without hair removal with data preparation, are shown in Fig. 1, and the second after hair removal with data preparation, as shown in Fig. 2.

3.5.4 The Framework of the Model

Figure 6 summarises the suggested Ensemble framework. We conducted two experiments with the framework. For Experiment 1, we trained three VGG16 models in parallel with three seed values, 42, 52, and 62, and input from Fig. 1. We then took the Ensemble soft average of the three models to get the final accuracy and classified them into seven skin cancer classes. For Experiment 2, a similar experiment was performed as in Experiment 1, but the difference is here: we have taken Fig. 2 as input to the experiment.

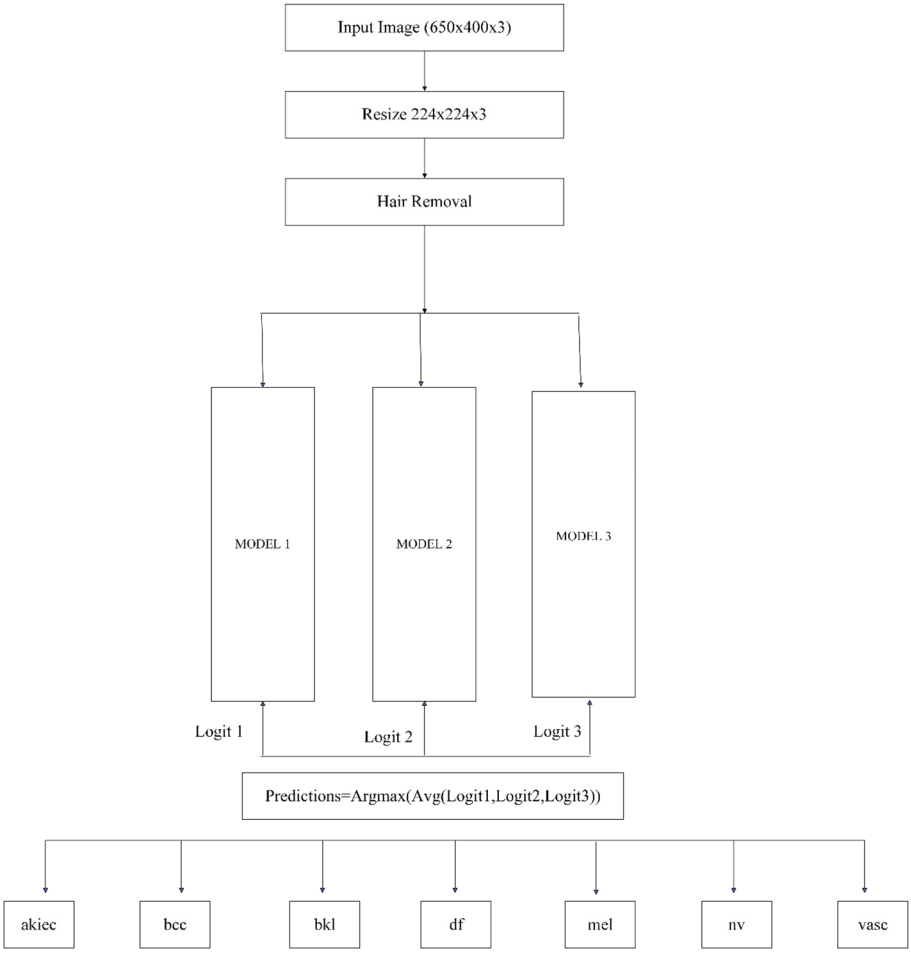


Fig. 6 Proposed Architecture

The Soft average takes the average of the predictions from three models i.e., Model 1, Model 2, and Model 3. Based on the overall prediction 828 test images are classified into seven classes.

4 Results and Discussion

4.1 Evaluation Metrics

The following indicators are taken into account for measuring performance to assess the suggested model.

a) Accuracy: Out of all the samples, precision counts how many predictions were accurate.

$$Accuracy = \frac{True\ Positive\ +\ True\ Negative}{Total\ Number\ of\ Samples}$$

b) Precision: The number of positive class forecasts that fall into the positive class is measured by precision.

$$Precision = \frac{True\ Positive}{True\ Positive\ +\ False\ Positive}$$

c) Recall: The number of accurate positive class predictions among all the real positive cases in the dataset is measured by recall.

$$Recall = \frac{True\ Positive}{True\ Positive\ +\ False\ Negative}$$

d) F1-Score: The F1-Score measure, the harmonic mean of the precision and recall from the preceding two tests, yields a single score representing the combined measure of these two tests.

$$F1 - Score = \frac{Precision * Recall}{Precision + Recall}$$

e) Support: Support is the quantity of images each class uses to determine evaluation measures.
f) Macro Accuracy: Macro averages or macro accuracy treat all classes equally by computing the standard for each class independently and then averaging the results.

The proposed strategy uses a T4 GPU high RAM (google colab) to train the network and is developed in Python. The hyperparameters used in our model include a learning rate of 0.01, no of epochs = 300, including early stopping criteria with a minimum delta of 0.001, patience = 65, and batch size = 50 as given in Table 3. The total parameters in our network are 60,217,671, with trainable parameters: 60,209,223 and non-trainable parameters: 8448. The activation function used is Relu and in the last layer softmax is used. The kernel of size 3 × 3 and stride of 1 used in the current study.

The standard measure used by Keras to characterize how well the model performed during the various training periods is termed accuracy. Table 4 describes the classification report of model 1, model 2, model 3, and the ensemble model of the previous three without

Table 3 Hyperparameters used by the model

Split Ratio	80:20
Kernal size	3×3
Stride	1
Learning Rate	0.01
Batch Size	50
Number Of Epochs	300
Minimum delta	0.001
Patience	65
Trainable Parameters	60,209,223
Non-Trainable Parameters	8448

hair removal images of the HAM10000 dataset. From the table, it can be seen that the Ensemble model gives us better results than individual models. Figure 7 shows the confusion matrix of the ensemble model where the diagonal of the matrix gives us the true predictions.

The individual class predictions of Model 1, Model 2, Model 3, and Ensemble Model without hair removal are graphically represented in Fig. 8.

Figure 9 compares the accuracy of individual model's vs Ensemble models without hair removal. From the figure, it can be seen that the ensemble model has higher accuracy than the individual ones.

Table 5 describes the classification report of model 1, model 2, model 3, and the ensemble model of the previous three with hair removal images of the HAM10000 dataset. From the table, it can be seen that the Ensemble model gives us better results than individual models. Figure 10 shows the confusion matrix of the ensemble model with hair removal.

The individual class predictions of Model 1, Model 2, Model 3, and Ensemble Model after hair removal are graphically represented in Fig. 11.

Figure 12 compares the accuracy of individual model's vs Ensemble models after hair removal. From the figure, it can be seen that the ensemble model has higher accuracy than the individual ones.

4.2 Accuracy and loss per epoch

The training and validation attributes are frequently examined using accuracy vs. epoch and loss vs. epoch curves, which must accurately match a deep learning model to fit a dataset. This analogy shows that our models are learning rather than retaining the information and are equally effective with hidden and visible data. Alternatively, significant variances across the training and validation attributes indicate that the algorithm is not over-fitted because it is optimized for the training dataset.

Therefore, the accuracy of a well-built model should rise with the epoch that experiences the fewest fluctuations, while the loss per epoch tends to follow training and validation negatively. Our three VGG16 models (i.e., Model 1, Model 2, and Model 3) hold these characteristics, as shown by their accuracy vs. epoch and loss vs. epoch curves.

The training accuracy and loss graphs of three models after hair removal are plotted in Fig. 13; they show different learning patterns for the same model architecture by varying initializations but approaching similar minima and almost the same accuracy and loss. With the same Learning rate = 0.01 for all three models but varying initialization, model1

Table 4 Classification Report of Ensemble Model without Hair Removal Technique

	Precision	Recall	F1-Score	Support
Classification Report Model 1				
Alkie	0.41	0.30	0.35	23
Bcc	0.55	0.65	0.60	26
Bkl	0.62	0.48	0.54	66
Df	0.40	0.33	0.36	6
Mel	0.41	0.47	0.44	34
Nv	0.93	0.95	0.94	663
Vasc	0.88	0.70	0.78	10
Accuracy			0.86	828
Macro Accuracy	0.60	0.56	0.57	828
Weighted Average	0.86	0.86	0.86	828
Classification Report Model 2				
Alkie	0.67	0.52	0.59	23
Bcc	0.70	0.54	0.61	26
Bkl	0.53	0.52	0.52	66
Df	0.33	0.67	0.44	6
Mel	0.49	0.50	0.49	34
Nv	0.94	0.95	0.95	663
Vasc	0.80	0.80	0.80	10
Accuracy			**0.87**	828
Macro Accuracy	0.64	0.64	0.63	828
Weighted Average	0.87	0.87	0.87	828
Classification Report Model 3				
Alkie	0.71	0.22	0.33	23
Bcc	0.53	0.62	0.57	26
Bkl	0.53	0.61	0.56	66
Df	0.50	0.50	0.50	6
Mel	0.49	0.56	0.52	34
Nv	0.94	0.94	0.94	663
Vasc	1.00	0.70	0.82	10
Accuracy			**0.86**	828
Macro Accuracy	0.67	0.59	0.61	828
Weighted Average	0.87	0.86	0.86	0.86
Classification Report Ensemble Model				
Alkie	0.67	0.35	0.46	23
Bcc	0.64	0.62	0.63	26
Bkl	0.58	0.59	0.59	66
Df	0.67	0.67	0.67	6
Mel	0.55	0.47	0.51	34
Nv	0.94	0.96	0.95	663
Vasc	0.89	0.80	0.84	10
Accuracy			**0.88**	828
Macro Accuracy	0.71	0.64	0.66	828
Weighted Average	0.88	0.88	0.88	828

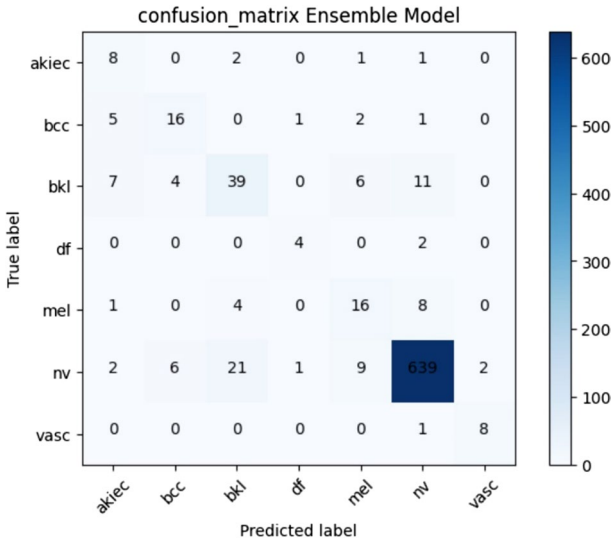


Fig. 7 Confusion matrix Ensemble model without hair removal Technique

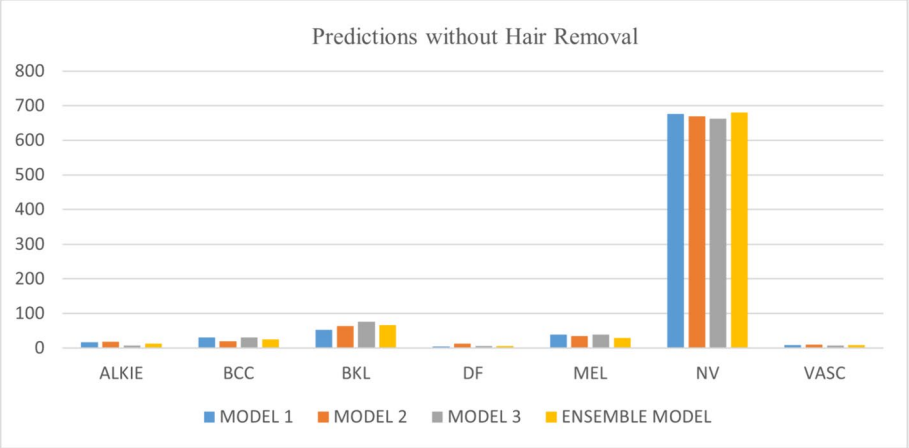


Fig. 8 Predictions of classes vs Models (without hair removal)

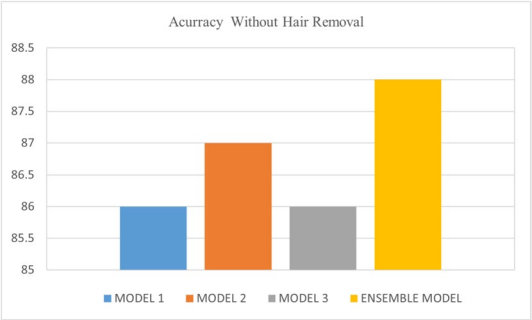


Fig. 9 Comparison of Accuracies individual models vs Ensemble Model (Without Hair Removal)

Table 5 Classification Report of Ensemble Model with Hair Removal Technique

	Precision	Recall	F1-Score	Support
Classification Report Model 1				
Alkie	0.61	0.48	0.54	23
Bcc	0.76	0.62	0.68	26
Bkl	0.53	0.56	0.54	66
Df	0.67	0.33	0.44	6
Mel	0.41	0.53	0.46	34
Nv	0.95	0.95	0.95	663
Vasc	1.00	0.60	0.75	10
Accuracy			**0.87**	828
Macro Accuracy	0.70	0.58	0.62	828
Weighted Average	0.88	0.87	0.87	828
Classification Report Model 2				
Alkie	0.59	0.43	0.50	23
Bcc	0.68	0.58	0.62	26
Bkl	0.52	0.73	0.61	66
Df	0.50	0.17	0.25	6
Mel	0.42	0.38	0.40	34
Nv	0.95	0.94	0.95	663
Vasc	1.00	0.70	0.82	10
Accuracy			**0.87**	828
Macro Accuracy	0.67	0.56	0.59	828
Weighted Average	0.88	0.87	0.87	828
Classification Report Model 3				
Alkie	0.67	0.26	0.38	23
Bcc	0.61	0.73	0.67	26
Bkl	0.51	0.53	0.52	66
Df	0.50	0.33	0.40	6
Mel	0.47	0.44	0.45	34
Nv	0.94	0.95	0.95	663
Vasc	0.57	0.80	0.67	10
Accuracy			**0.86**	828
Macro Accuracy	0.61	0.58	0.58	828
Weighted Average	0.86	0.86	0.86	828
Classification Report Ensemble Model				
Alkie	0.64	0.39	0.49	23
Bcc	0.76	0.73	0.75	26
Bkl	0.57	0.68	0.62	66
Df	1.00	0.33	0.50	6
Mel	0.46	0.47	0.46	34
Nv	0.95	0.96	0.96	663
Vasc	0.88	0.70	0.78	10
Accuracy			**0.89**	828
Macro Accuracy	0.75	0.61	0.65	828
Weighted Average	0.89	0.89	0.88	828

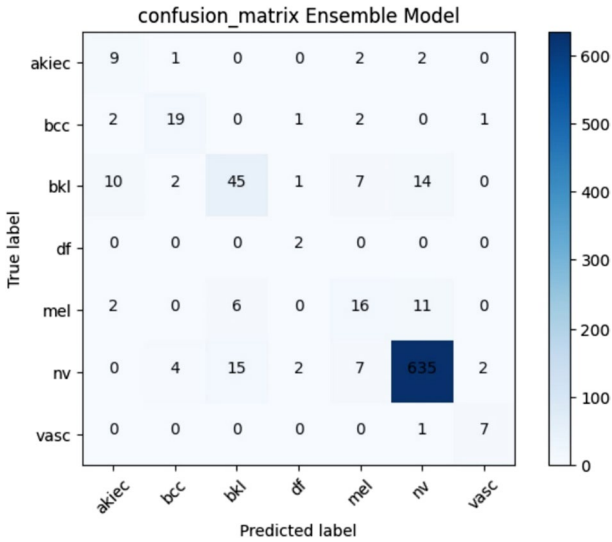


Fig. 10 Confusion matrix Ensemble model after the hair removal process

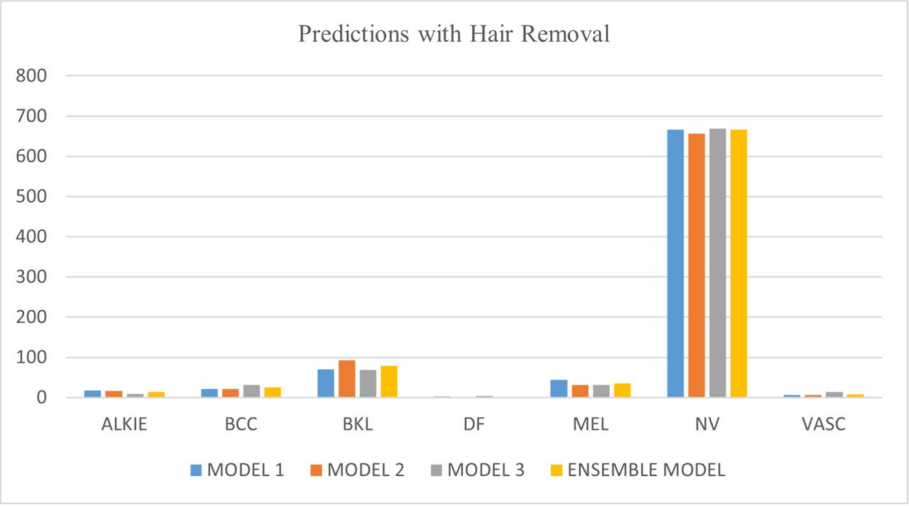


Fig. 11 Predictions of classes vs Models (with hair removal)

saturates at epoch 98, model2 saturates at epoch 113, and model 3 saturates at epoch 104. The individual models disagree with each other, so the Ensemble model performs better.

Table 6 compares the proposed model with previous models. It demonstrates how improved deep learning models and emergent image processing methods have raised our model's overall

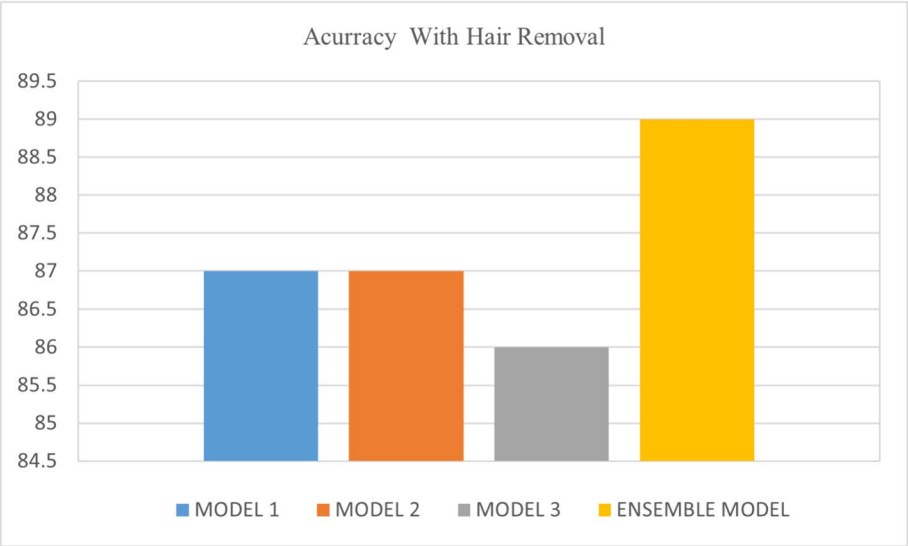
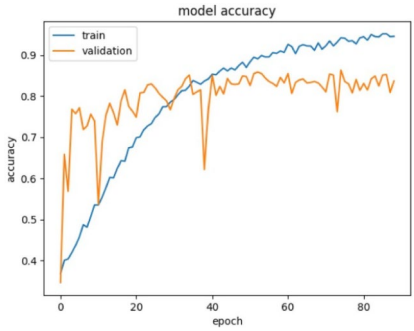


Fig. 12 Comparison of Accuracies individual models vs Ensemble Model (With Hair Removal)

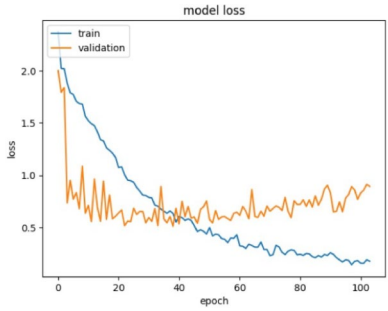
test accuracy. Without hair removal and Ensemble Technique, the accuracy of our model is 86%. After applying Hair removal and Ensemble techniques, the accuracy of our method is 89%. The suggested multi-class skin cancer classification algorithm outperforms the other current approaches by all measures.

5 Conclusion

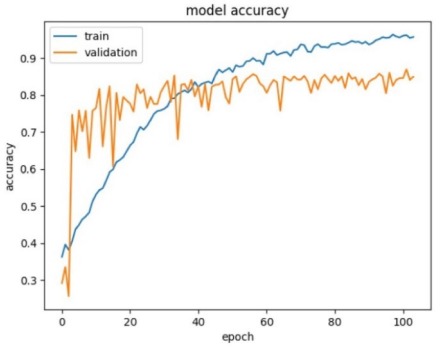
Skin cancer is difficult for dermatologists to detect in its early stages. In recent years, there has been a notable rise in the adoption of DL for supervised and unsupervised applications. Classifying skin cancer is an essential domain of exploration. The dataset comes from MNIST: HAM10000, which includes seven distinct categories of skin lesion images and has an overall size of 10,015. This research presents a DL architecture for multi-class skin cancer classification based on Ensemble VGG16 and the hair removal concept. The present investigation builds an ensemble network in which the predictions from three VGG16 sub-models are aggregated and supplied to a meta-learner to determine the final skin cancer class prediction. Here, the overall accuracy is 88% with the Ensemble model before the hair removal technique and 89% after the hair removal technique is obtained. An open-access HAM10000 image dataset is used to train and test the model with a ratio of 80:20. Test results further reveal that our model efficiently detects seven types of skin cancer using the hair removal technique



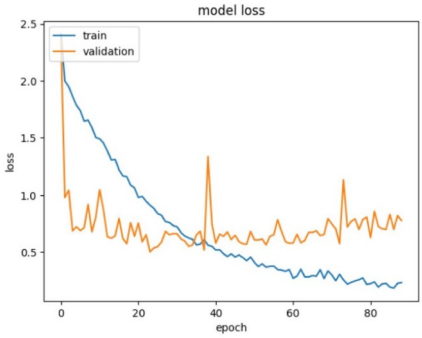
a) Training Accuracy vs Epoch Model 1 (Hair Removal)



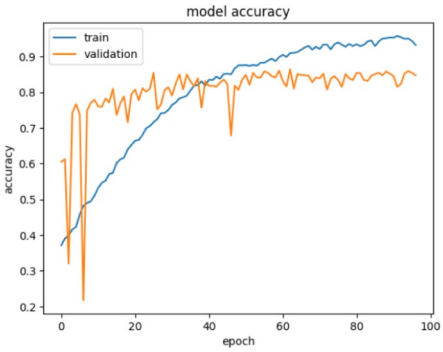
b) Loss vs Epoch Model 1 (Hair Removal)



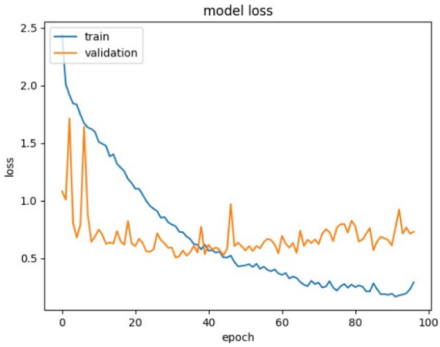
c) Training Accuracy vs Epoch Model 2 (Hair Removal)



d) Loss vs Epoch Model 2 (Hair Removal)



e) Training Accuracy vs Epoch Model 3 (Hair Removal)



f) Loss vs Epoch Model 3 (Hair Removal)

Fig. 13 Accuracy loss Graphs of three models after hair removal Technique

Table 6 Comparison of the proposed model with state-of-the-art models

Work	Accuracy	Precision	Recall	F1-Score
Wu J et al. [29]	85.7%	0.756	0.816	-
Farooq et al. [50]	86%	0.8	0.89	0.84
Wang et al. [56]	86%	-	-	0.86
Khan et al. [55]	81.8%	-	-	0.81
Alfi et al. [43]	88%	-	-	-
Xinyuan et al. [41]	87%	88.24	87.5	0.879
Ours (without hair removal and Ensemble)	**86%**	**0.86**	**0.86**	**0.86**
Ours (before hair removal but with ensemble)	**88%**	**0.88**	**0.88**	**0.88**
Ours (after hair removal and Ensemble)	**89%**	**0.89**	**0.89**	**0.89**

and ensemble model, demonstrating excellent explaining qualities. The ensemble model is used differently to increase accuracy but still needs improvement for real skin cancer applications. We also intend to improve prediction outcomes in the future by processing images and using cutting-edge DL networks.

Data Availability My manuscript has no associated data.

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